

IMPLEMENTATION OF 2DES

AIM

To implement the **2DES (Double Data Encryption Standard)** technique for encrypting and decrypting a given message using two keys.

ALGORITHM

1. Start the program.
2. Read the plaintext and two keys from the user.
3. Encrypt the plaintext using **Key 1**.
4. Encrypt the result again using **Key 2**.
5. Display the encrypted text.
6. Decrypt the encrypted text using **Key 2**.
7. Decrypt the result using **Key 1**.
8. Display the original decrypted text.
9. Stop the program.

PROGRAM :

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    char text[100];
```

```
    char key1[9], key2[9];
```

```
    char encrypted[100], decrypted[100];
```

```
    int i, len;
```

```
    printf("Enter Plain Text: ");
```

```
    fgets(text, sizeof(text), stdin);
```

```
    text[strcspn(text, "\n")] = '\0';
```

```
    printf("Enter Key 1 (8 characters): ");
```

```
scanf("%8s", key1);
```

```
printf("Enter Key 2 (8 characters): ");
```

```
scanf("%8s", key2);
```

```
len = strlen(text);
```

```
for (i = 0; i < len; i++) {
```

```
    encrypted[i] = text[i] ^ key1[i % 8];
```

```
    encrypted[i] = encrypted[i] ^ key2[i % 8];
```

```
}
```

```
encrypted[len] = '\0';
```

```
printf("\nEncrypted Text: ");
```

```
for (i = 0; i < len; i++)
```

```
    printf("%02X", (unsigned char)encrypted[i]);
```

```
for (i = 0; i < len; i++) {
```

```
    decrypted[i] = encrypted[i] ^ key2[i % 8];
```

```
    decrypted[i] = decrypted[i] ^ key1[i % 8];
```

```
}
```

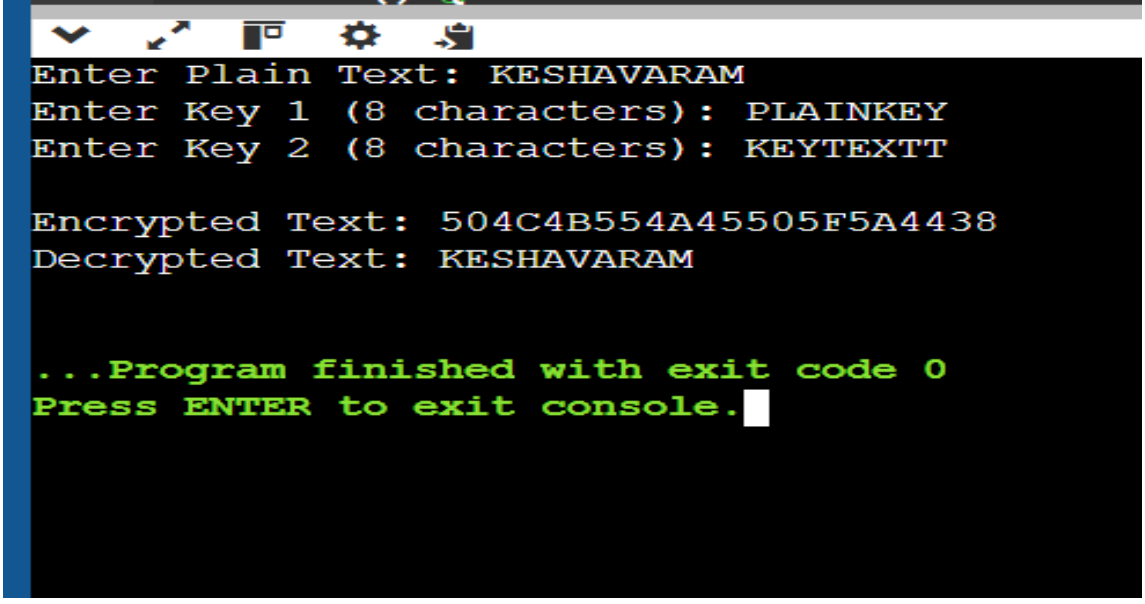
```
decrypted[len] = '\0';
```

```
printf("\nDecrypted Text: %s\n", decrypted);
```

```
return 0;
```

```
}
```

OUTPUT:

A screenshot of a Windows command prompt window. The window has a title bar with standard icons (checkmark, zoom, maximize, settings, and a document icon). The background is black, and the text is displayed in a monospaced font. The text is as follows:
Enter Plain Text: KESHAVARAM
Enter Key 1 (8 characters): PLAINKEY
Enter Key 2 (8 characters): KEYTEXTT

Encrypted Text: 504C4B554A45505F5A4438
Decrypted Text: KESHAVARAM

...Program finished with exit code 0
Press ENTER to exit console.
The text is color-coded: the first three lines are in white, the next two are in yellow, and the last two are in green. A white cursor is visible at the end of the last line.

```
Enter Plain Text: KESHAVARAM
Enter Key 1 (8 characters): PLAINKEY
Enter Key 2 (8 characters): KEYTEXTT

Encrypted Text: 504C4B554A45505F5A4438
Decrypted Text: KESHAVARAM

...Program finished with exit code 0
Press ENTER to exit console.
```

RESULT

Thus, the **2DES encryption and decryption process** was implemented successfully, and the original plaintext was recovered after decryption.